COURSE DESCRIPTION FORM: CS-4110 High Performance Computing with GPUs

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

PROGRAM TO BE EVALUATED	BS-CS – **Fall-2025**

Course Description

Course Code	CS-4110
Course Title	High Performance Computing with GPUs
Credit Hours	3+0
Prerequisites by Course(s) and Topics	
Grading Policy	Absolute grading
Policy about missed assessment items in the course	Retake of missed assessment items (other than sessional/ final exam) will not be held. For a missed sessional / final exam, an exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decides the exam retake/ pretake cases.
Course Plagiarism Policy	Plagiarism in project or sessional/ final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	***100% Theory*** Assessment Items **Assessment Item** / **Number** / **Weight (%)** Assignments / >2 / 8 Complex Computing Problem / 1 / 12 Sessional 1 / 1 / 15 Sessional 2 / 1 / 15 Quizzes / >4 / 10 Final Exam / 1 / 40
Course Instructors	Dr. Imran Ashraf
Lab Instructors (if any)	
Course	Dr. Imran Ashraf

Assessment Items

Assessment Item	Number	Weight (%)
Assignments	>2	8
Complex Computing Problem	1	12
Sessional 1	1	15
Sessional 2	1	15
Quizzes	>4	10
Final Exam	1	40

Coordinator	
URL (if any)	
Current Catalog Description	Growing demand for computing in the domains of Artificial Intelligence (AI), Financial computing, Quantum Computing, Engineering simulations etc. is being satisfied mainly by GPU-based HPC systems. This course will teach the fundamentals needed to utilize the ever-increasing power of the accelerators, especially GPUs in an HPC system. The course will start with an architectural overview of modern HPC, especially GPU-based heterogeneous system, focusing on its computing power versus data movement needs. The course will cover a high level (pragma-based) GPU programming approach (Directive-based programming) as well utilization of existing libraries to quickly prototype an HPC application. Breadth of the course will cover lower-level approaches based on CUDA and OpenCL programming language for finer grained computationally intensive tasks. A particular attention will be given on performance tuning and techniques to overcome common data movement bottlenecks and patterns.
Textbook (or **Laboratory Manual** for Laboratory Courses)	David B. Kirk, Wen-mei W. Hwu, "Programming Massively Parallel Processors A Hands-on Approach", 3rd Edition (November 24, 2016), Elsevier.
Reference Material	1. Robert Robey, Yuliana Zamora , "Parallel and High Performance Computing", 1st edition (July 2021), Manning Publications 2. Aaftab Munshi, Benedict Gaster, Timothy G. Mattson, James Fung, Dan Ginsburg, OpenCL Programming Guide, 1st Edition, Publisher: Addison-Wesley Professional; 1st edition (July 23, 2011) 3. Nvidia eBook, CUDA C++ Programming Guide, https://docs.nvidia.com/cuda/cuda-c-programming-guide/index.html 4. Nvidia eBook, HPC for the Age of AI and Cloud Computing, https://www.nvidia.com/en-us/lp/high-performance-computing/hpc-ai-cloud-computing-ebook/
Course Learning Outcomes	**A. Course Learning Outcomes (CLOs)** After completion of the course, the students shall be able to: 1. **Describe** the terms related to HPC systems, GPU architecture and GPU programming models. 2. **Identify** application hotspots based on the application profile using realistic workload. 3. **Develop** data-parallel solutions using appropriate programming model. 4. **Analyze** performance of an application running on an HPC system to improve compute and memory performance.

PLO 1	Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems
PLO 2	Problem Analysis	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.

PLO 3	Design/Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PLO 4	Investigation & Experimentation	Conduct investigation of complex computing problems using research-based knowledge and research-based methods.
PLO 5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.
PLO 6	Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.
PLO 7	Environment and Sustainability	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.
PLO 8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.
PLO 9	Individual and Teamwork	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PLO 10	Communication	Communicate effectively on complex computing activities with the computing community and with society at large
PLO 11	Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member of a team.
PLO 12	Lifelong Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.

C. Mapping of CLOs to PLOs
(CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)

		PLOs											
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
CLOs	1	✓											✓
	2		✓	✓	✓	✓				✓			
	3			✓									
	4				✓	✓				✓			

Topics covered in the course with number of lectures on each topic (assume 15-week instruction and 3 contact hours per week)	**1. Topics to be covered:**

List of Topics	No. of Weeks	Contact Hours	CLO(s)
Introduction to HPC systems, applications, trends, taxonomy	**1**	**3**	**1**
Application profiling, dependence analysis, performance analysis, Amdahl's law, HPC system theoretical performance estimation, compute power and memory bandwidth computation	**2**	**6**	**2**
GPU Architecture: Streaming multiprocessor, Thread scheduler, Memory hierarchy	**1**	**3**	**1**
GPU Programming Fundamentals • Basics of CUDA program, device and host code, data types • Blocks, threads and wraps, CPU-GPU data transfer, Kernel launch configuration and device occupancy • Device synchronization, Unified memory model	**3**	**9**	**3**
Memory and Communication Optimization	**1**	**3**	**4**
Directive-based GPU Programming	**1**	**3**	**3**
OpenCL-based GPU Programming	**1**	**3**	**3**
Acceleration of Image Processing and Engineering simulations on GPUs	**1**	**3**	**3**
Acceleration of ML/DL Applications on GPUs	**1**	**3**	**3**
CUDA Streams	**1**	**3**	**3**
PyCUDA, Mult-GPU HPC, Intel OneAPI, OpenMP Offload Model, Performance Portability	**1**	**3**	**1**

Future directions, course mapping to HPC job market, Project presentations	**1**	**3**	**1, 2, 3, 4**
Total	**15**	**45**	

Laboratory Projects/ Experiments Done in the Course	NA			
Class Time Spent (in percentage)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethical Issues**
	20	25	50	5
Oral and Written Communications	Every student is required to submit at least 1 written reports/implementation of the semester project and to make oral presentation/demo of typically 10 minute's duration.			